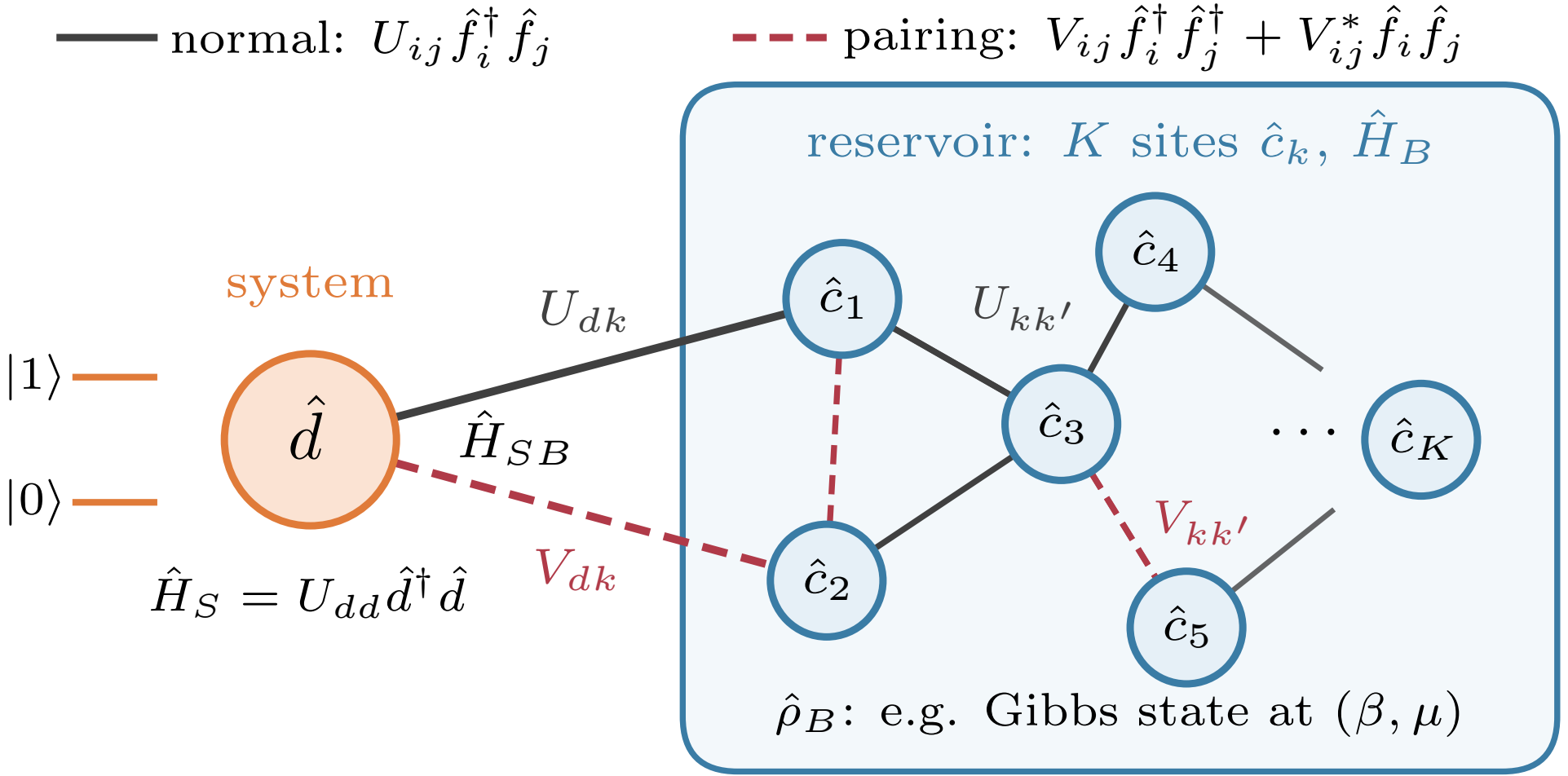


$$\hat{\rho}(0) = \hat{\rho}_S(0) \otimes \hat{\rho}_B, [\hat{\rho}_B, \hat{\mathcal{P}}] = 0$$